Lecture 07: Higher order linear non-homogeneous ODE with constant coefficients

Rule 3: (i) If $y_p = \frac{1}{f(D^2)} \sin bx$ or $\cos bx$, then

$$y_p = \frac{1}{f(-b^2)} \sin bx \text{ or } \cos bx \quad \text{if } f(-a^2) \neq 0.$$

(Successful case)

Examples: Solve $(D^2+4)y = \sin 3x$

A: Given ODE is

$$(D^2+4)y = \sin 3x \quad \text{--- (1)}$$

A.E. $m^2+4=0 \Rightarrow m = \pm 2i$

$$y_c = e^0[A\cos 2x + B\sin 2x]$$
$$= A\cos 2x + B\sin 2x$$

(1) $\Rightarrow$ $y_p = \frac{1}{D^2+4}\sin 3x$

$$= \frac{1}{-3^2+4}\sin 3x = -\frac{1}{5}\sin 3x$$

$\therefore$ G.S. $y = y_c + y_p = A\cos 2x + B\sin 2x - \frac{1}{5}\sin 3x$

(ii) Failure Case: $y_p = \frac{1}{f(-b^2)} \sin bx$ or $\cos bx$

and if $f(-b^2) = 0$ then ...

Example: Solve $(D^2+b^2)y = (\cos bx$ or $\sin bx)$

$$y_p = \frac{1}{D^2+b^2}(\sin bx \text{ or } \cos bx)$$

$$= \frac{x}{2}\int(\sin bx \text{ or } \cos bx)\,dx$$

$$=$$

Example 3: Solve $(D^2+1)y = \sin x$

Ans: Given $(D^2+1)y = \sin x$ — (1)

A.E. $m^2+1=0 \Rightarrow m = \pm i$

$$y_c = A\cos x + B\sin x$$

(1) $\Rightarrow$ $y_p = \frac{1}{D^2+1}\sin x$ [Here $b=1$, $D^2 = -1^2$ Pm, $Dr = 0$]

$$= \frac{x}{2}\int \sin x\,dx$$

$$= \frac{x}{2}(-\cos x) = -\frac{x}{2}\cos x$$

$\therefore$ G.S $y = y_c + y_p = A\cos x + B\sin x - \frac{x}{2}\cos x$.

Example 4: Solve $(D^2-3D+2)y = \sin 3x$

Ans: Given $(D^2-3D+2)y = \sin 3x$ —— (1)

A.E. $m^2-3m+2=0$; $m=1,2$

$$y_c = c_1e^{x}+c_2e^{2x}$$

(1) $\Rightarrow$ $y_p = \dfrac{1}{D^2-3D+2}\sin 3x$ [Here $b=3$]

$$\Rightarrow y_p = \frac{1}{-3^2-3D+2}\sin 3x = \frac{1}{-7-3D}\sin 3x$$

$$\Rightarrow y_p = \frac{1}{-(7+3D)}\sin 3x = \frac{7-3D}{-(7+3D)(7-3D)}\sin 3x$$

$$\Rightarrow y_p = -\frac{7-3D}{49-9D^2}\sin 3x = \frac{3D-7}{49-9(-3^2)}\sin 3x$$

$$\Rightarrow y_p = \frac{3\cos 3x.(3)-7\sin 3x}{49+81}$$

$$= \frac{1}{130}(9\cos 3x - 7\sin 3x)$$

$\therefore$ G.S. $y = y_c + y_p$

$$= c_1e^{x}+c_2e^{2x}+\frac{1}{130}(9\cos 3x-7\sin 3x)$$

Example 5: Solve $(D^2-3D+4)y = \cos(4x+5)$

Hints: Here $b=4$

Rule 4: If $y_p = \frac{1}{f(D)} e^{ax} M(x)$ then

$$y_p = e^{ax} \frac{}{f(D+a)} M(x)$$ [$M(x)$ = Polynomial of x or $\sin bx$ or $\cos bx$]

Example 6: Solve $(D^2-2D+1)y = e^x \sin x$

Ans: Given $(D^2-2D+1)y = e^x \sin x$ — (1)

A.E. $m^2-2m+1=0 \Rightarrow m=1,1$

$$y_c = c_1 e^x + c_2 x e^x$$

(1) $\Rightarrow y_p = \frac{1}{D^2-2D+1} e^x \sin x$ (Here $a=1$)

$$= e^x \frac{1}{(D+1)^2-2(D+1)+1} \sin x$$

$$= e^x \frac{1}{D^2+\cancel{2D}+1-\cancel{2D}-\cancel{2}+1} \sin x$$

$$= e^x \frac{1}{D^2} \sin x$$ [Now we can apply Rule 3]

$$= e^x \frac{1}{D} \int \sin x \, dx$$

$$= e^x \frac{1}{D} (-\cos x)$$

$$= e^x (-\sin x)$$

$$= -e^x \sin x$$

Rule 3 using:

$$y_p = e^x \frac{1}{-1^2} \sin x$$

$$= -e^x \sin x$$

G.S. $y = y_c + y_p$

$$= c_1 e^x + c_2 x e^x - e^x \sin x$$

Ans.

Example of: Solve $(D^2-4D+4)y = x^2 e^{2x}$

Ans: Given $(D^2-4D+4)y = x^2 e^{2x}$ — (1)

A.E. $m^2-4m+4=0 \therefore m=2,2$

$$y_c = c_1 e^{2x} + c_2 x e^{2x}$$

(1) $\Rightarrow$ $y_p = \dfrac{1}{D^2-4D+4} e^{2x} x^2$ [Here $a=2$]

$$= e^{2x} \frac{1}{(D+2)^2-4(D+2)+4} x^2$$

$$= e^{2x} \frac{1}{D^2+\cancel{4D}+\cancel{4}-\cancel{4D}-\cancel{4}+\cancel{4}} x^2$$

$$= e^{2x} \frac{1}{D^2} x^2$$

$$= e^{2x} \frac{1}{D} \int x^2 \, dx = e^{2x} \frac{1}{D}\left(\frac{x^3}{3}\right)$$

$$= e^{2x} \frac{x^4}{12}$$

$\therefore$ G.S. $y = c_1 e^{2x} + c_2 x e^{2x} + \frac{1}{12} x^4 e^{2x}$ Ans

Home task: (MKC book) Page 109

Example: 4.8, 4.9, 4.10, 4.11.

Zill's book (Ch 5) p-146 Example 8: Solve $y''' + y'' = e^x \cos x$ — (1)

(6)

A: A.E. $m^3 + m^2 = 0 \Rightarrow m^2(m+1) = 0$

$\therefore m = 0, 0, -1$

$$\therefore y_c = c_1 e^0 + c_2 x e^0 + c_3 e^{-x}$$

(1) $\Rightarrow$ $$y_p = \frac{1}{D^3 + D^2} e^x \cos x = e^x \frac{1}{(D+1)^3 + (D+1)^2} \cos x$$

$$\Rightarrow y_p = e^x \frac{1}{D^3 + 3D^2 + 3D + 1 + D^2 + 2D + 1} \cos x$$

$$= e^x \frac{1}{D^3 + 4D^2 + 5D + 2} \cos x$$ (We can apply Rule 3)

$$= e^x \frac{1}{D \cdot D^2 + 4D^2 + 5D + 2} \cos x = e^x \frac{1}{(-1)^2 D + 4(-1)^2 + 5D + 2} \cos x$$

$$y_p = e^x \frac{1}{-D - 4 + 5D + 2} \cos x = e^x \frac{1}{4D - 2} \cos x$$

$$y_p = e^x \frac{4D + 2}{(4D)^2 - 2^2} \cos x = e^x \frac{4D + 2}{16D^2 - 4} \cos x$$

$$y_p = e^x \frac{4(-\sin x) + 2\cos x}{16(-1)^2 - 4} = e^x \frac{-4\sin x + 2\cos x}{20}$$

$$y_p = \frac{1}{5} e^x \sin x - \frac{1}{10} e^x \cos x$$

A.S $$y = c_1 + c_2 x + c_3 e^{-x} - \frac{1}{10} e^x \cos x + \frac{1}{5} e^x \sin x.$$